

MOUNIB NASR

AI Software Engineer --- Multi-Agent Systems, RAG & Production ML/LLM Platforms

mounibnasr123@gmail.com | +216 50 239 112 | Tunisia
github.com/mounibnasr45 | linkedin.com/in/nasr | mounibnasr.onrender.com

Professional Summary

AI Software Engineer with hands-on, production experience building generative AI systems: multi-agent LangGraph orchestration, hybrid RAG pipelines (dense + lexical retrieval), and MCP tool servers, translated into testable, auditable agentic workflows via versioned prompts and Human-in-the-Loop validation. Comfortable across the full GenAI stack — prompt engineering, LLM agents, retrieval-augmented generation, and cloud AI platforms (Vertex AI, Gemini) — and use AI-assisted engineering tools (Claude Code, GitHub Copilot) as daily productivity multipliers.

Technical Skills

- **Languages:** Python, SQL, Java
- **Agentic AI:** LangGraph (Send, reflect/retry, checkpointing), MCP/FastMCP, A2A, LangChain, CrewAI, BeeAI
- **RAG & Retrieval:** Hybrid BM25 + dense (RRF), pgvector, Qdrant HNSW, Graphiti/Neo4j, semantic chunking
- **System Design:** Clean architecture, Repository/Factory/State machine patterns, microservices, async processing
- **Backend & Data:** FastAPI, AsyncIO, SQLAlchemy, Pydantic, PostgreSQL, MongoDB, Neo4j, Redis, MinIO/S3
- **Observability:** Langfuse, structured logging, golden-dataset regression, prompt-version tracking, HITL
- **DevOps & ML:** Docker(Compose), CI/CD, Prefect, GCP Vertex AI/Gemini, PyTorch, LoRA, YOLO, MLflow
- **AI-Assisted Engineering:** Claude Code (daily driver), GitHub Copilot, versioned prompts/agent configs

Experience

ASTRO Lab

AI Software Engineer — Final-Year Project

Feb 2026 – Present

Sousse, Tunisia

- **Developed** SkillBridge AI, a multi-tenant SaaS platform that identifies employee skill gaps, generates justified learning roadmaps, and computes 4 distinct business metrics via a native explainability engine.
- **Architected** a contract-first LangGraph pipeline separating semantic LLM classification from a deterministic scoring engine, incorporating parallel branches, bounded reflection loops, and Redis-backed checkpointing.
- **Engineered** a 10-service containerized infrastructure (FastAPI, PostgreSQL, Qdrant, Next.js 14) featuring Human-in-the-Loop validation and real-time SSE streaming for multilingual (FR/EN) PDF report generation.

Sky Light

AI Systems Architect — Freelance

Sep 2025 – Jan 2026

Remote

- **Reduced** competitive marketing analysis time from days to minutes by deploying an automated, multi-agent AI platform tailored for digital advertising and media buyers.
- **Orchestrated** autonomous agents using CrewAI to execute complex ads analysis, keyword research, and competitor discovery for on-demand, client-ready report generation.
- **Constructed** a scalable backend utilizing FastAPI, Prefect, and Docker, establishing a unified data schema across Similarweb, BuiltWith, Google Ads, and Meta Ad Library.

Drug-IT

AI Intern — R&D FormulAI Platform

Jun 2025 – Jul 2025

Remote

- **Formulated** the FormulAI platform, an integrated Graph RAG system designed to accelerate pharmaceutical formulation development by transforming unstructured scientific text into actionable insights.
- **Implemented** an intelligent ingestion pipeline utilizing Mistral OCR, Gemini embeddings, and Chonkie for semantic chunking (1,000-2,500 tokens), supported by a multi-level SHA256 caching system.
- **Programmed** temporal knowledge graphs using Neo4j and Graphiti to extract, track, and efficiently traverse multi-hop relationships between pharmaceutical entities over time.

Multiverse.ai

Intern — Computer Vision & ML in Production

Dec 2024 – May 2025

Remote

- **Deployed** computer vision models (ViT, YOLO) achieving 94% top-1 accuracy while processing 1,000+ images daily.
- **Managed** the ML lifecycle via MLflow and Docker, eliminating manual visual inspection bottlenecks across assembly workflows.

SeekMake

ML Intern — OCR & NLP Data Processing

Jul 2024 – Sep 2024

Remote

- **Streamlined** unstructured document extraction by creating automated OCR/NLP pipelines.
- **Slashed** manual annotation effort by 70% via an intuitive, self-serve data processing workflow for non-technical teams.

Selected Projects

[Tunisian Legal AI Assistant](#) — *LangGraph • PostgreSQL/pgvector • BM25 • FastAPI*

- **Pioneered** a hybrid RAG platform (BM25 + pgvector) delivering legal answers grounded in Tunisian legislation, achieving 98.2% hit@5 accuracy across benchmark evaluations.
- **Integrated** self-reflection checkpoints and automated CI regression testing over a 56-question evaluation harness to guarantee verifiable article citations.

[Medical Agentic System](#) — *CrewAI • GraphRAG (Neo4j) • FastAPI • LLMs*

- **Created** a medical AI assistant combining CrewAI and GraphRAG to provide secure, evidence-grounded clinical decision support.
- **Structured** an adaptive workflow routing complex queries to specialized clinical agents, maintaining strict conversational context and high-precision vector retrieval.

[Agentic Deep Research System](#) — *LangGraph • BeeAI • FastMCP • A2A Protocol*

- **Conceptualized** a fully autonomous research platform utilizing LangGraph and the A2A protocol for seamless coordination between planner, interviewer, and writer agents.
- **Incorporated** live web search and Wikipedia FastMCP tools with ReAct reasoning, enforcing a stringent quality-gate validation prior to generating final reports.

[Movie Recommender System](#) — *Matrix Factorization • SentenceTransformers • FastAPI*

- **Established** a hybrid recommendation engine fusing matrix factorization with content-based filtering, significantly mitigating new-user cold-start issues.
- **Containerized** a semantic search layer via SentenceTransformers and FastAPI using Docker Compose, optimizing natural-language movie discovery endpoints.

Independent Research — Hubness in Approximate Nearest-Neighbor Search — *HNSW indexing • Statistical Validation • Data Structures*

- **Investigated** hubness impact on approximate nearest-neighbor search, revealing a 5.3x variance in distance evaluations at matched recall levels for high-skew datasets.
- **Executed** a pre-registered experimental protocol ensuring statistical validity, reproducible HNSW indexing benchmarks, and transparent engineering decisions (ADRs).

Education & Languages

National Engineering Diploma — Software Engineering, Intelligent & Distributed Systems
ISIMM, University of Monastir

2021 – 2026

Languages: Arabic (Native) • English (B2) • French (B2)